EC 571: Digital VLSI Circuit Design

Flip-flops and Latches

Ajay Joshi

Acknowledgement - Some slides adapted from:
- J. Rabaey et al. Digital Integrated Circuits 2nd Ed.
- UC Berkeley EE141, UC Berkeley EE241, UC Berkeley EE251
- Stanford EE271

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Last lecture

- Last lecture
  - Domino Logic
- Today's lecture
  - Flip-flops and latches
- Reading
  - Chapter 7

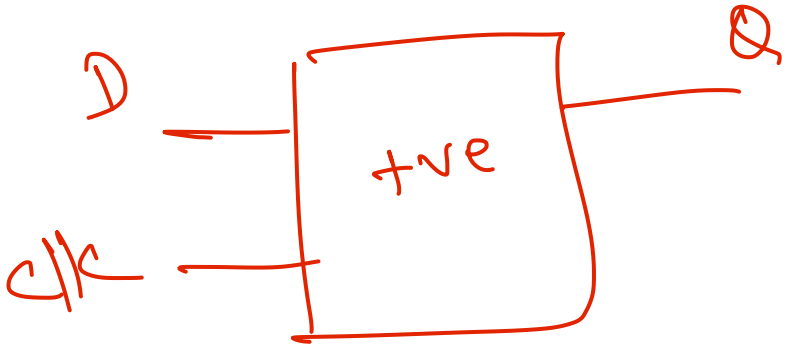
Latch

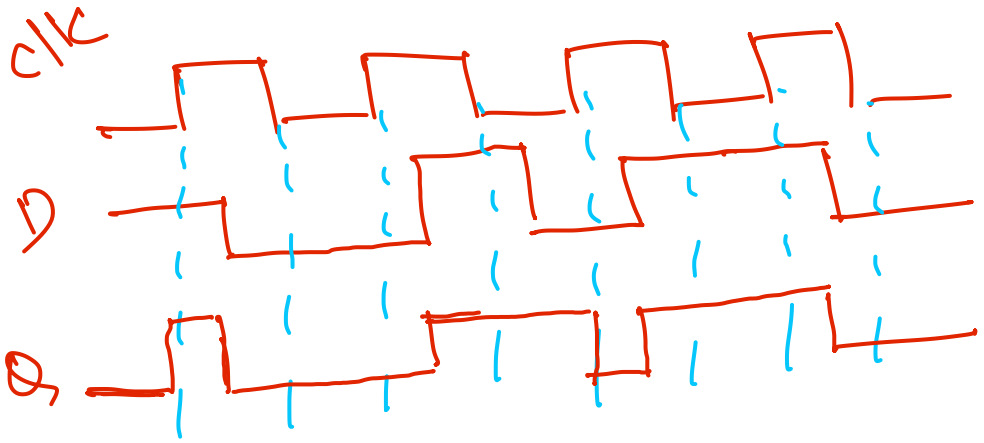
Level-triggered.

clk = 0 opaque

clk = 1 transparent-

+ve level triggered



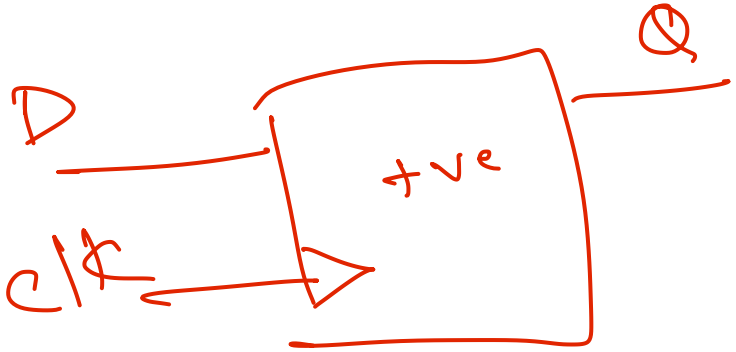


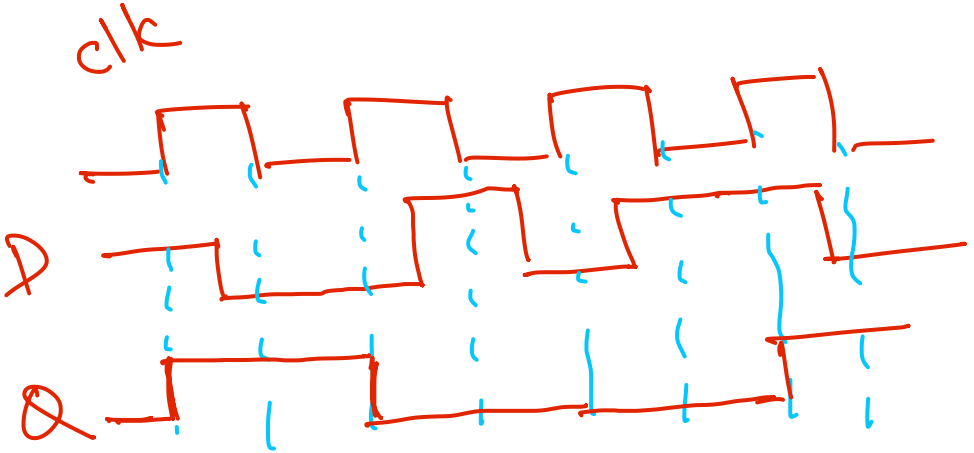
Flip flop

Edge-triggered

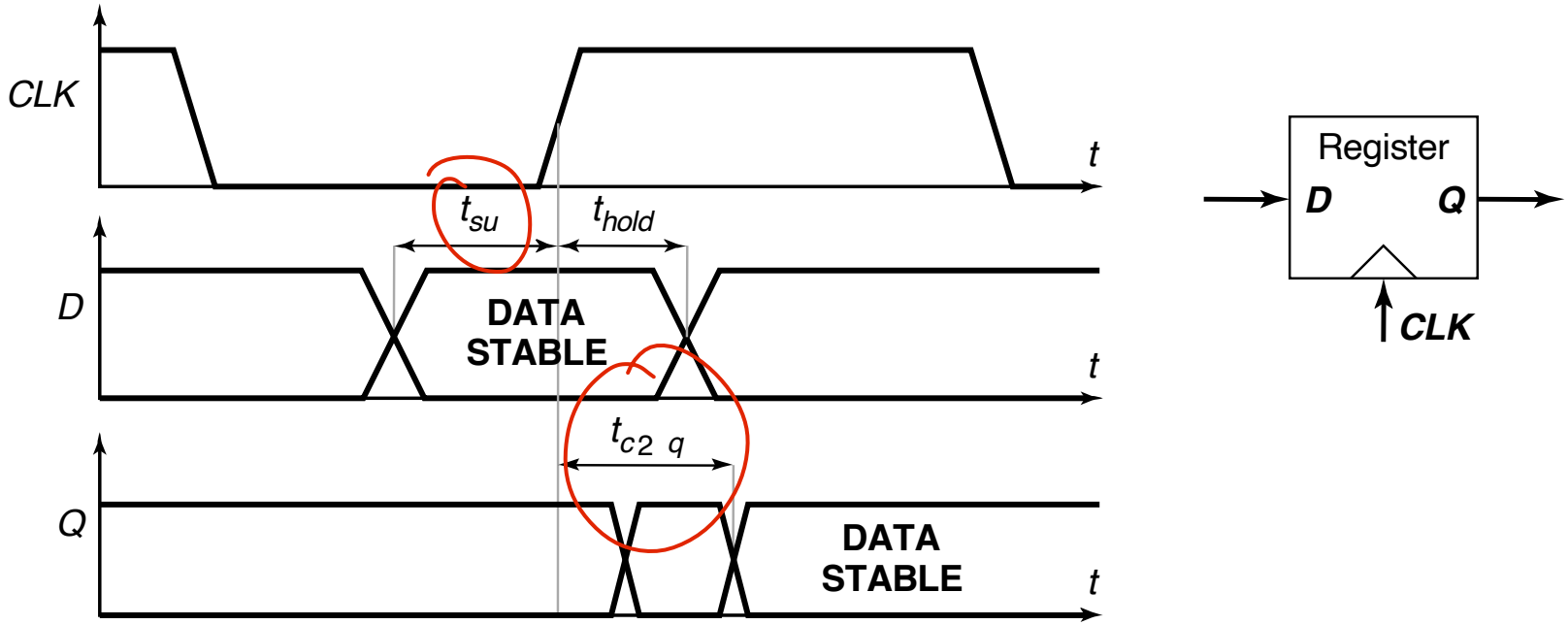
Stores new data on the +ve clk edge

+ve edge-triggered

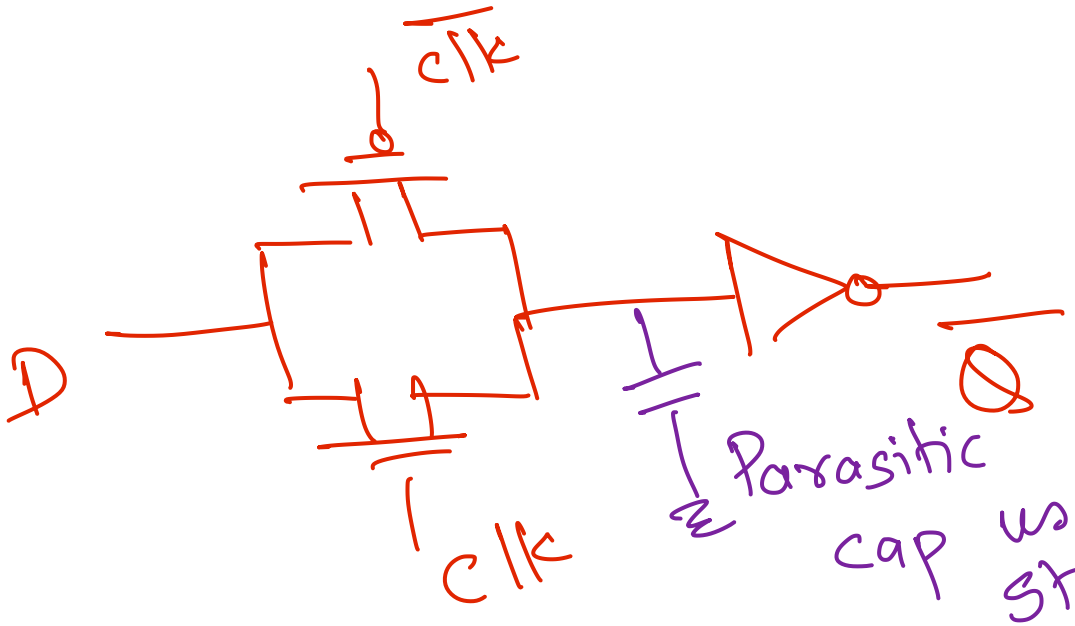




Timing Definitions



Storage mechanism

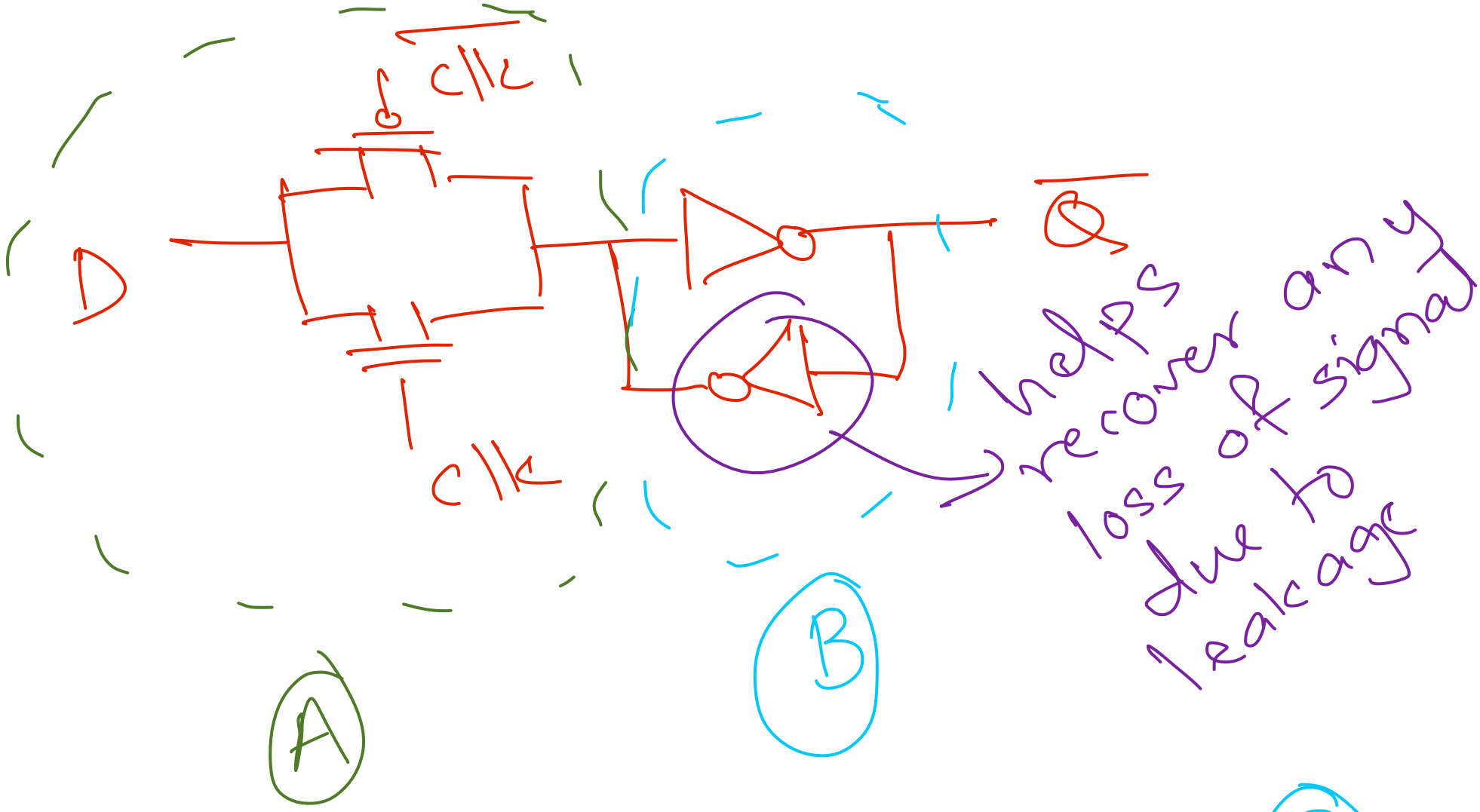


Dynamic latch.

cap used to store data.

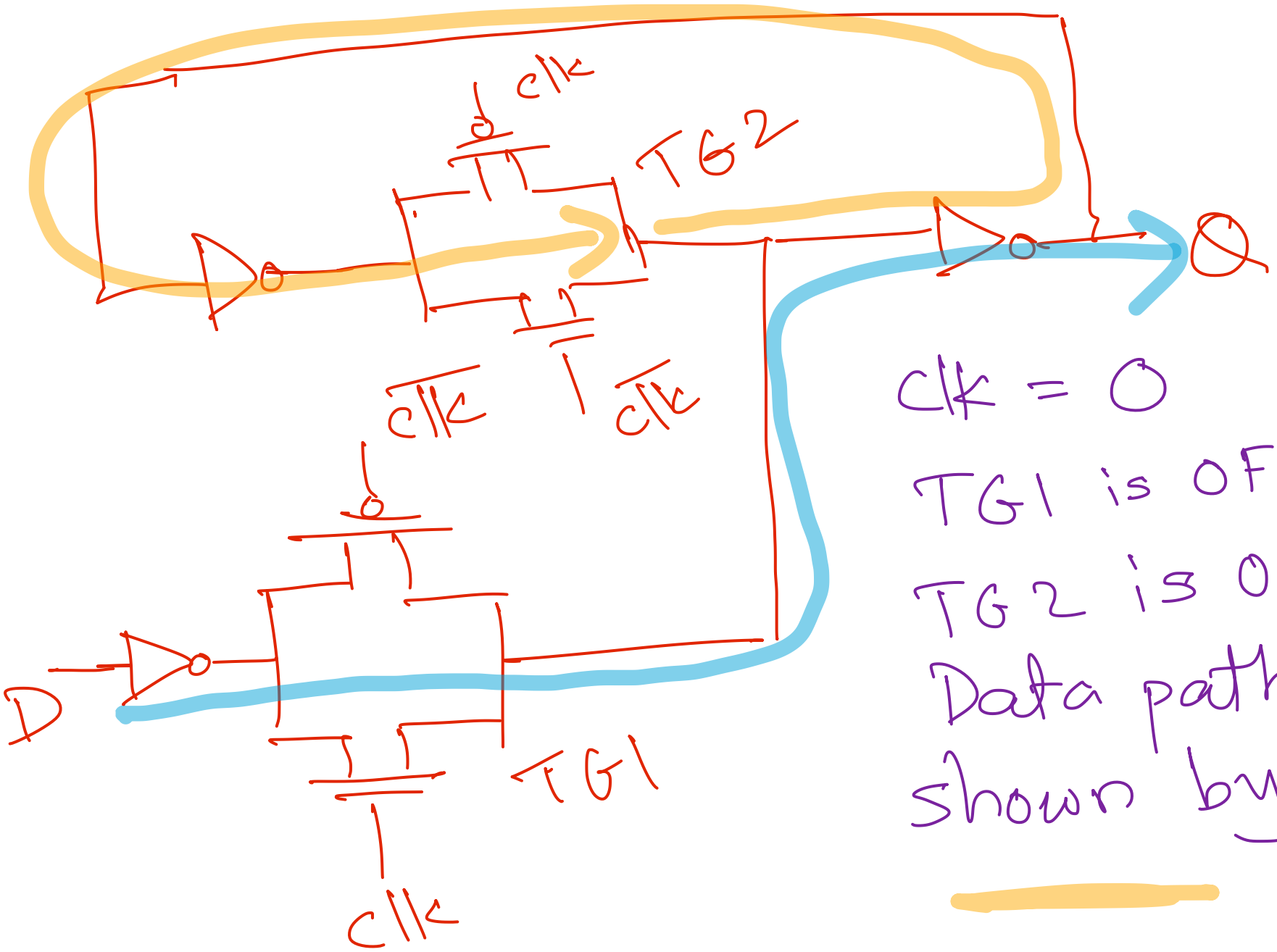
↓

Cap will leak when TG is OFF



A is able to overpower B to write data in the latch

Alternate way to address the issue



clk = 1

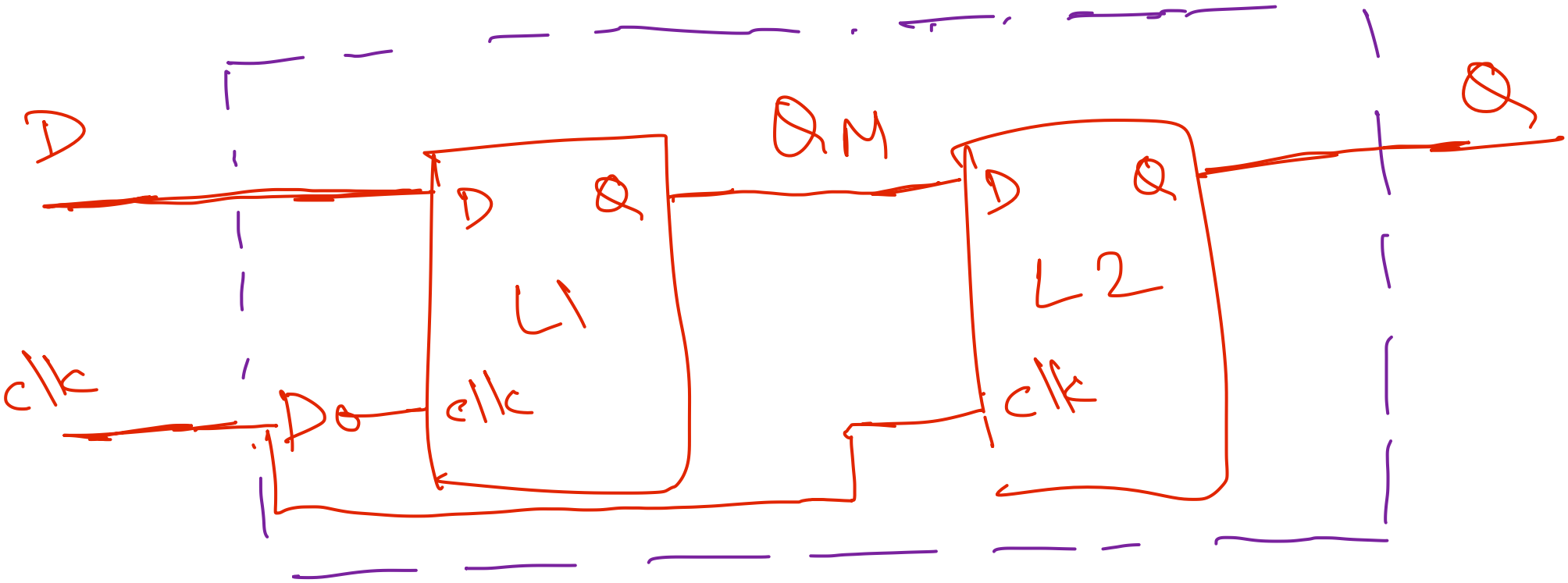
TG1 is ON

TG2 is OFF

Data path shown by

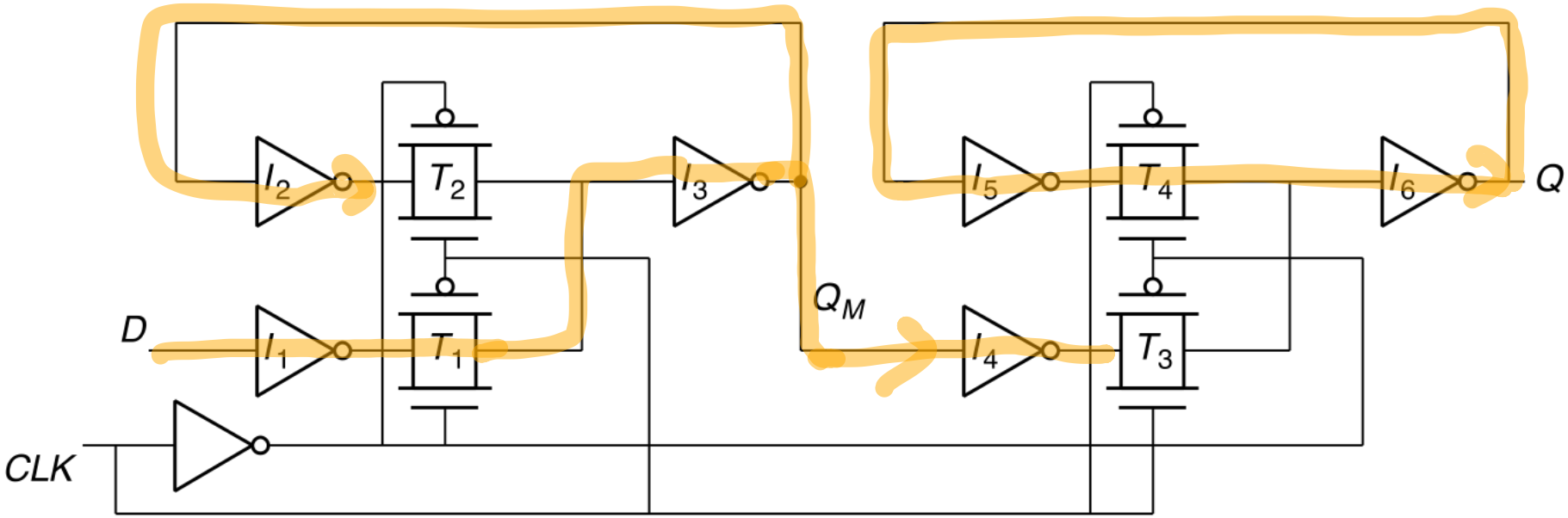
Positive - level triggered latch.

Design a +ve-edge triggered FF using two +ve-level triggered latches



Edge-Triggered Register

Multiplexer-based latch pair



clk = 0 path shown by —

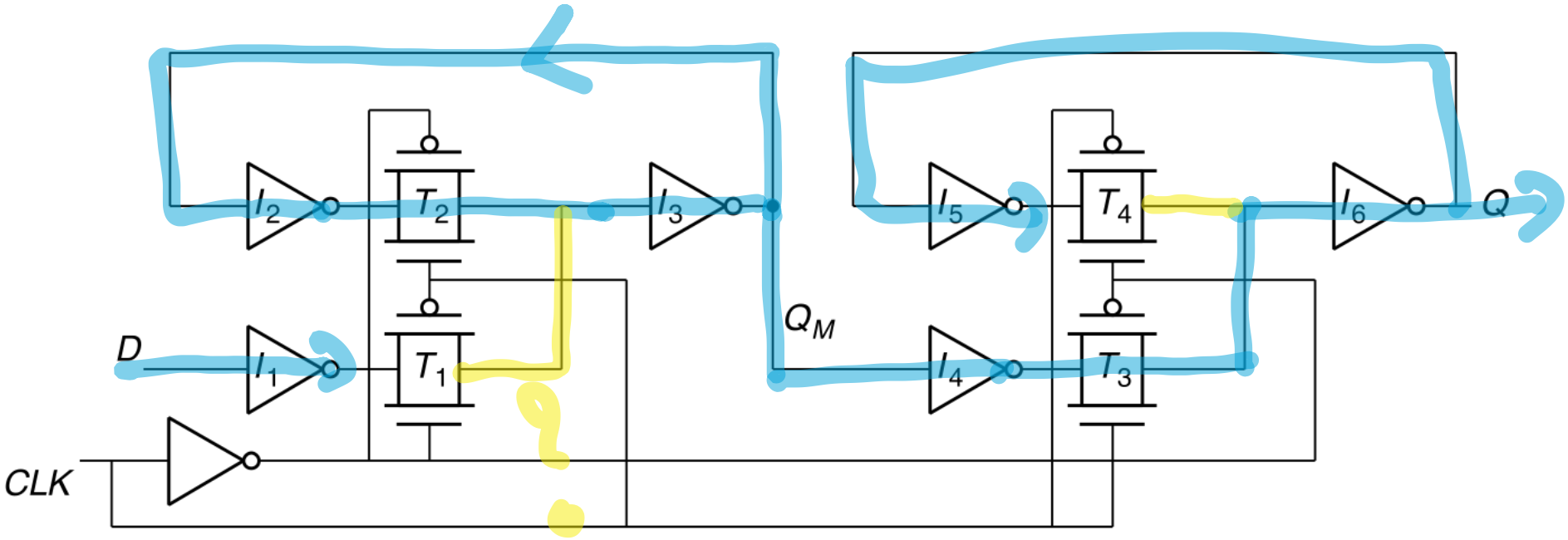
T1 ON T3 OFF

T2 OFF T4 ON

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Edge-Triggered Register

Multiplexer-based latch pair



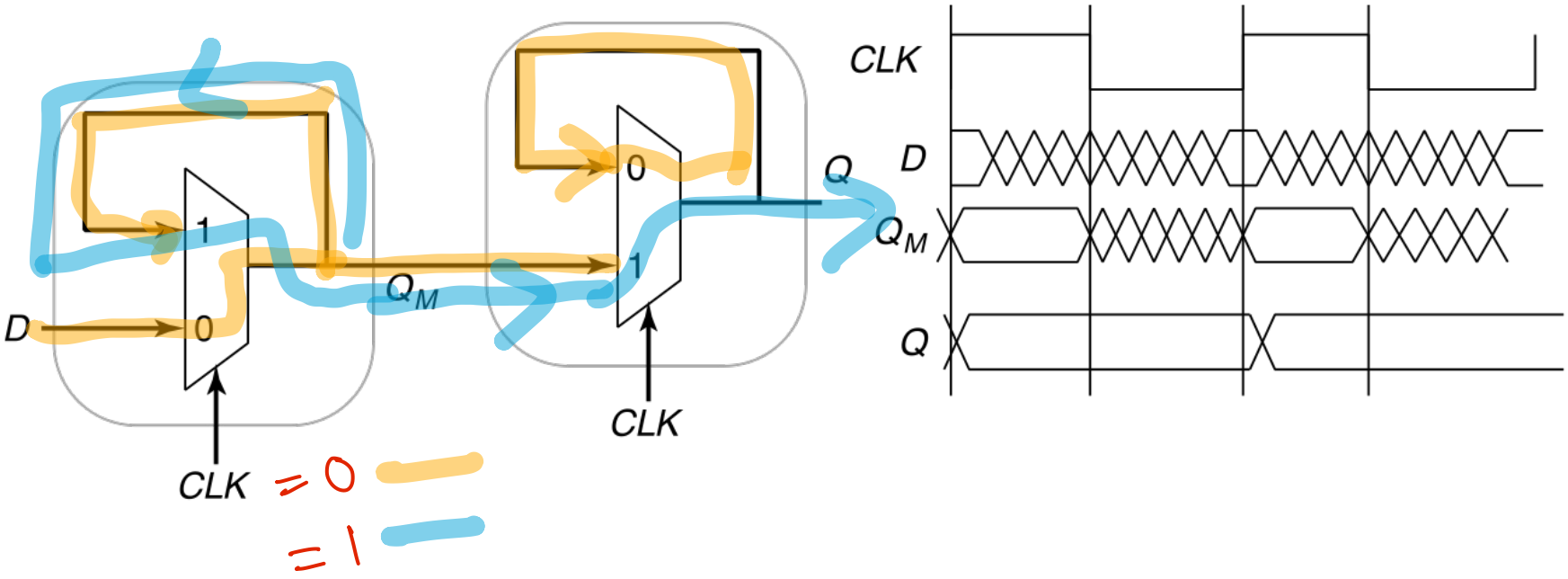
clk = 1 path shown by —

T1 OFF T3 ON

T2 ON T4 OFF

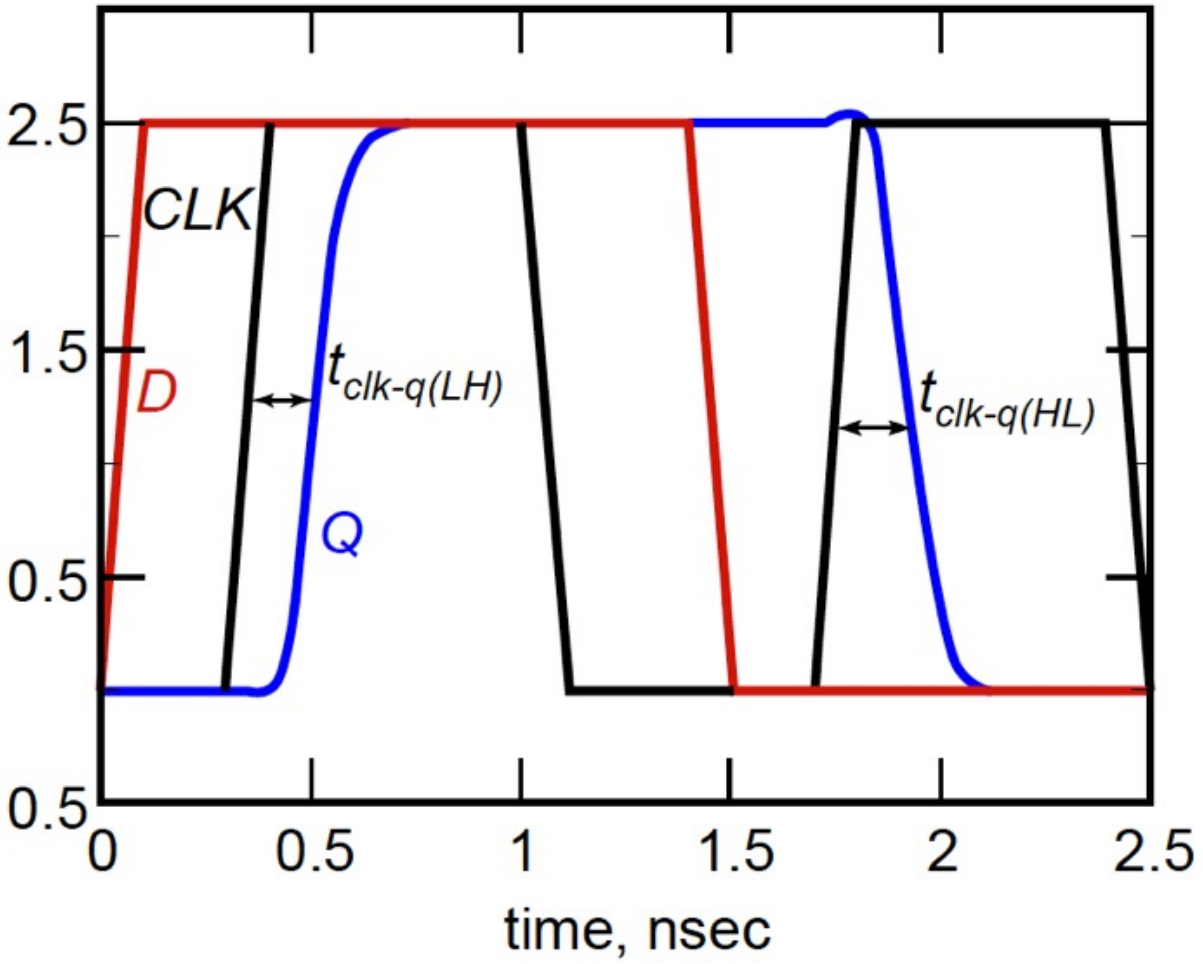
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Edge-Triggered Register



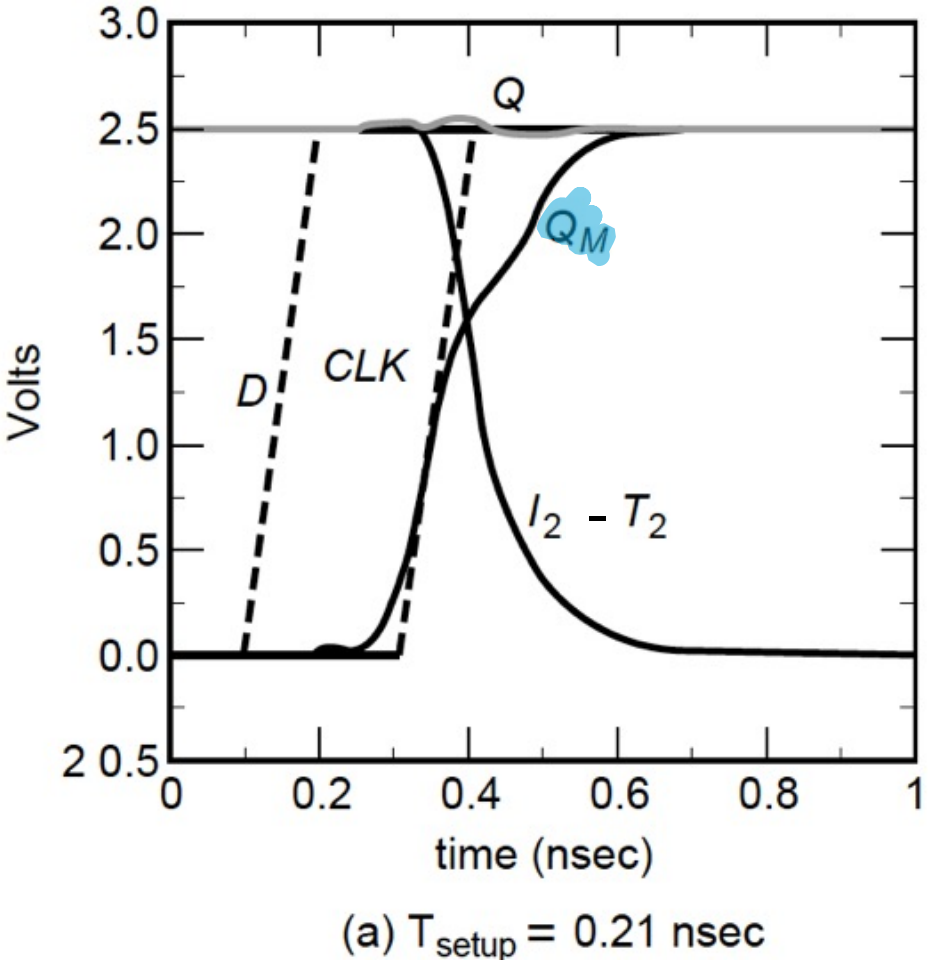
Two opposite latches trigger on edge

Clk-Q Delay

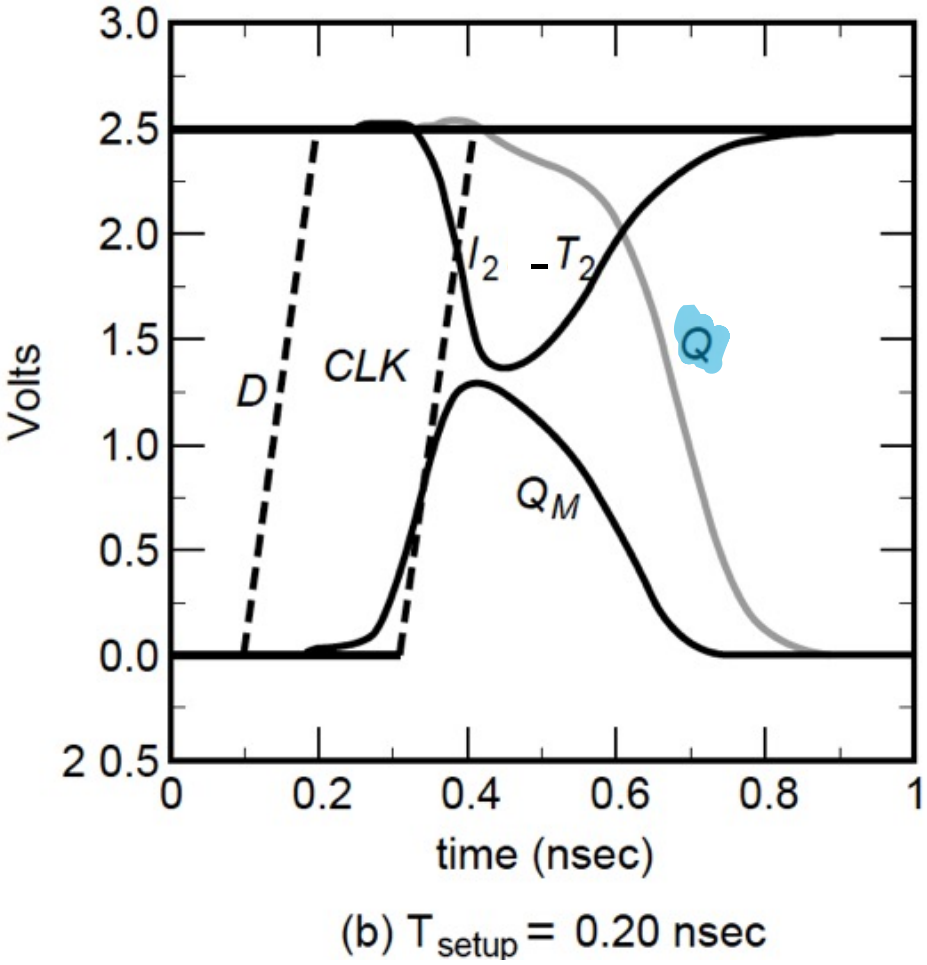


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Setup Time



(a) T_{setup} = 0.21 nsec

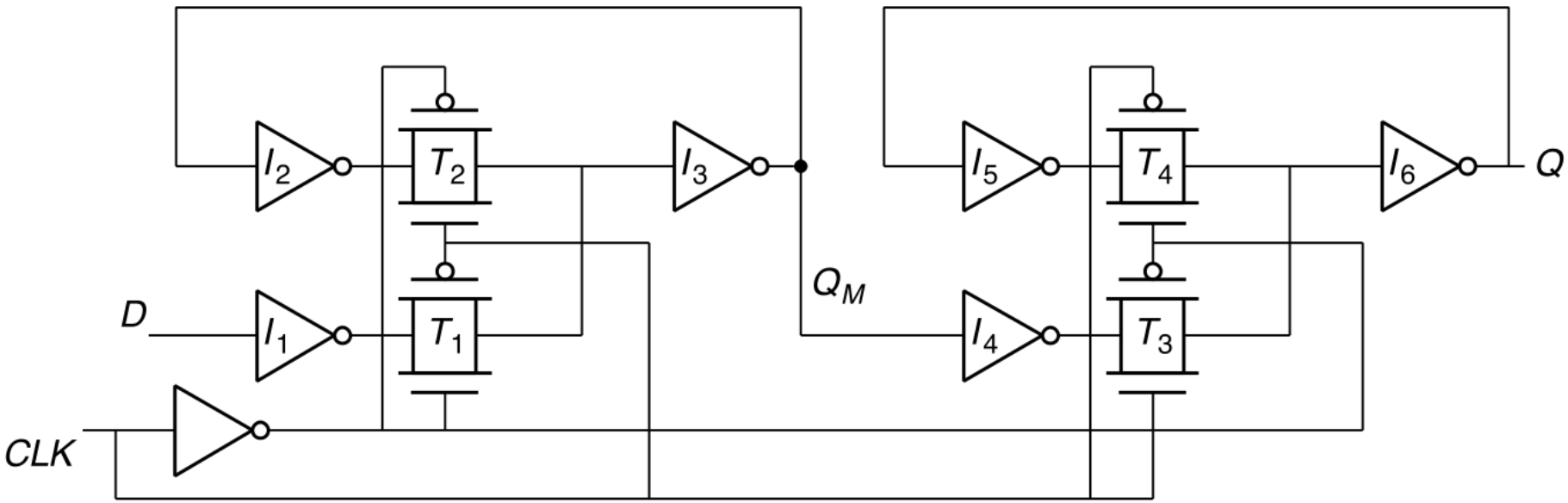


(b) T_{setup} = 0.20 nsec

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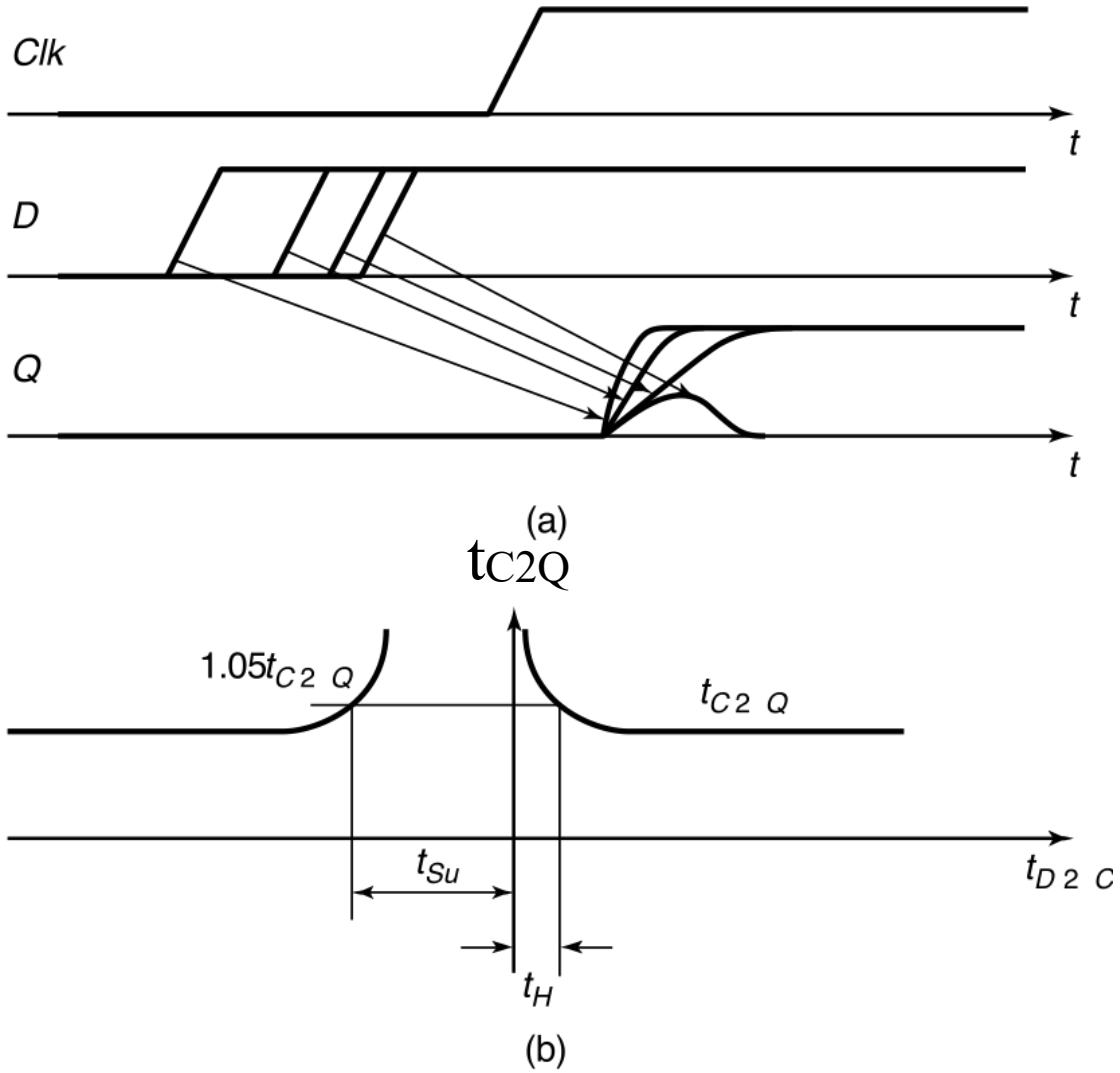
Edge-Triggered Register

Multiplexer-based latch pair



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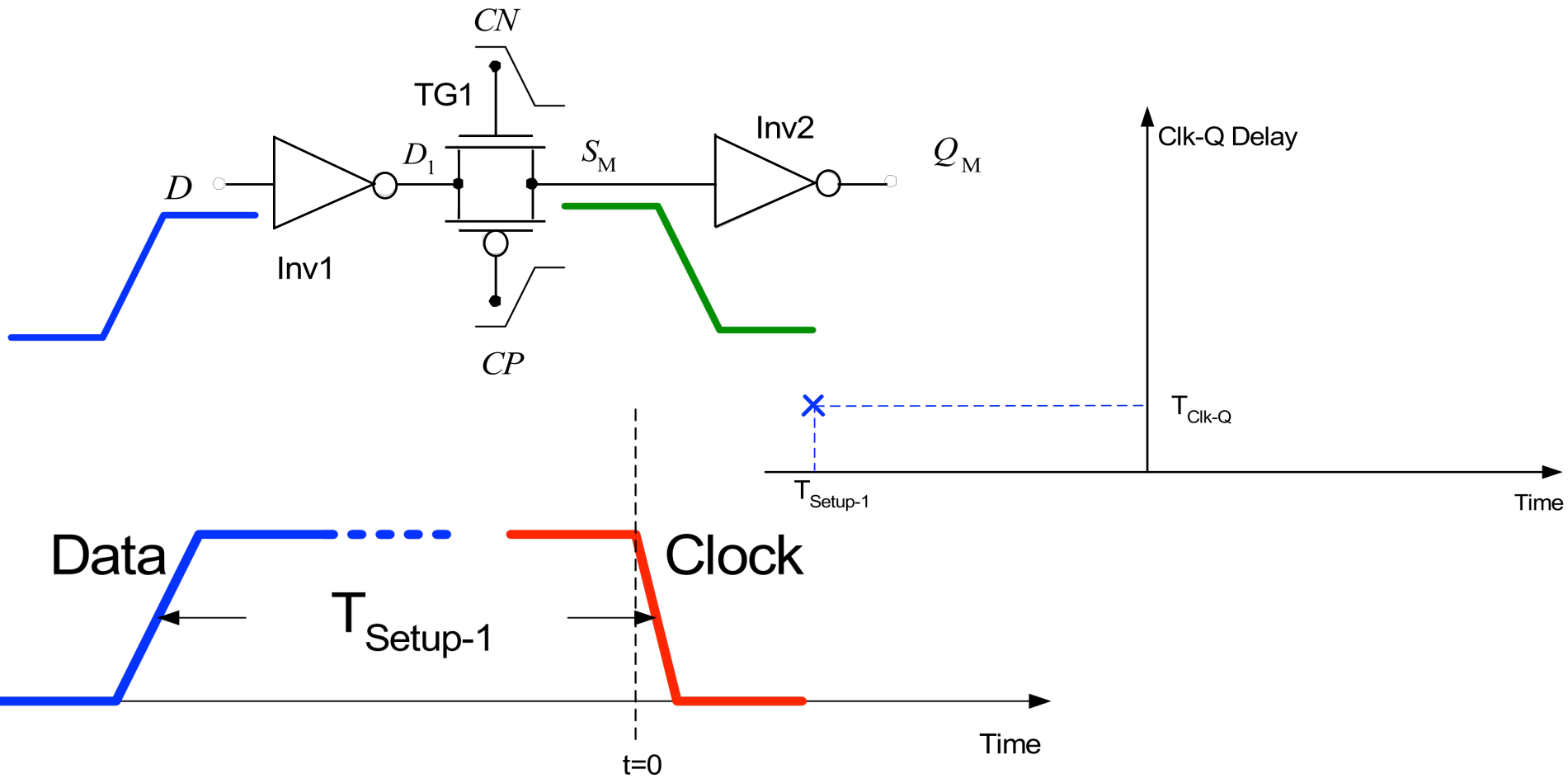
More Precise Setup Time



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Setup/Hold Time Illustrations

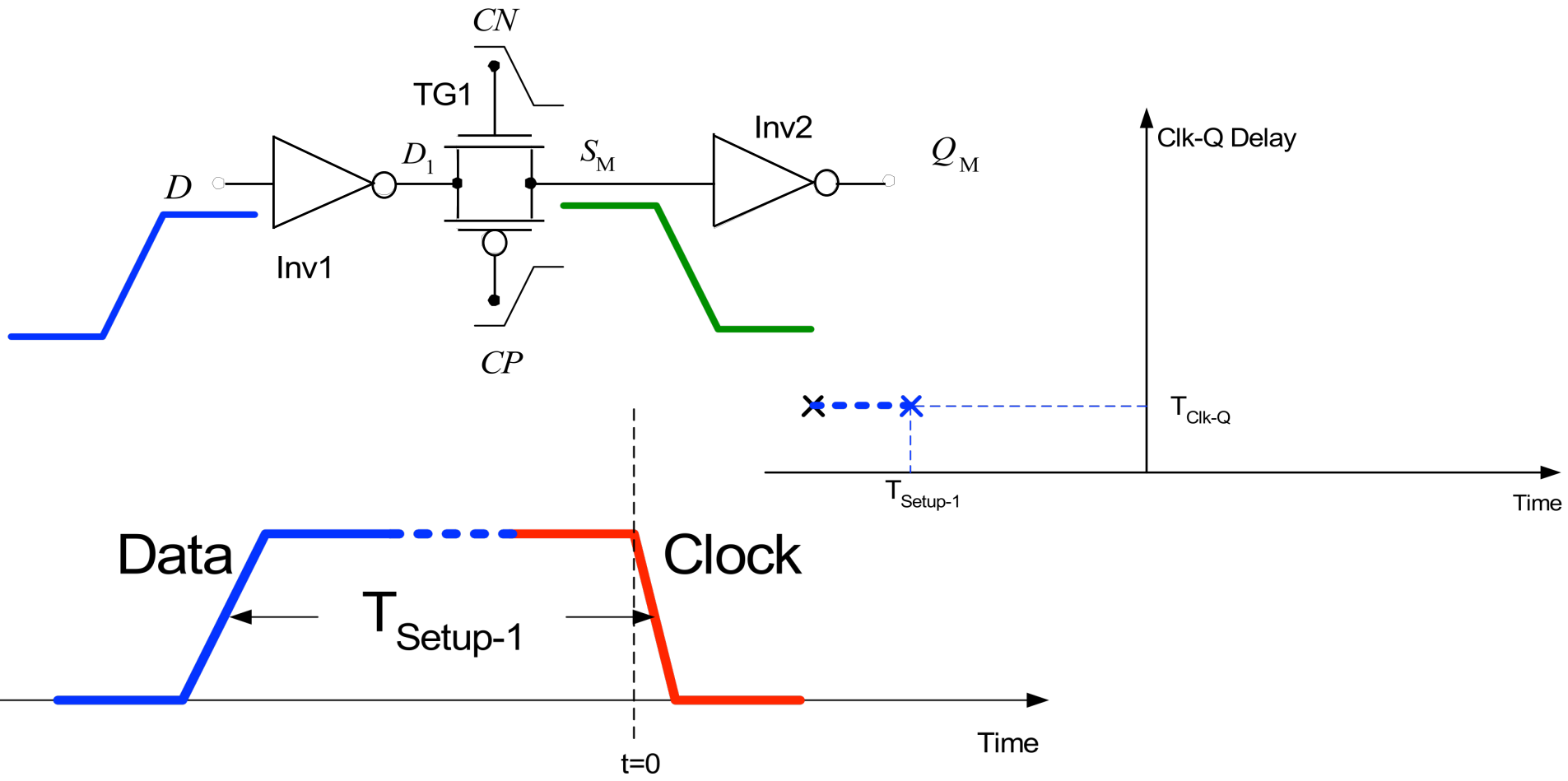
Circuit before clock arrival (Setup-1 case)



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Setup/Hold Time Illustrations

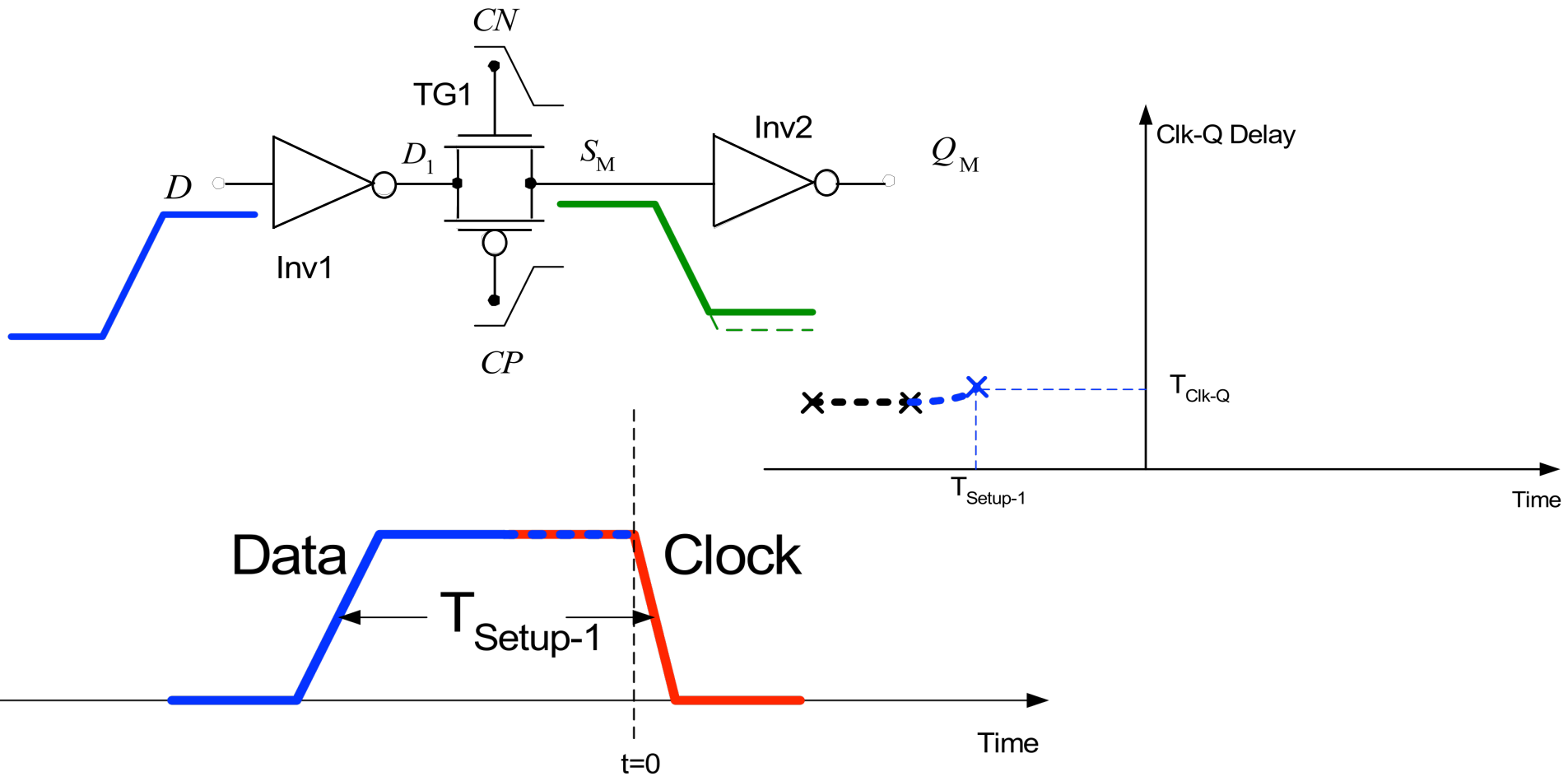
Circuit before clock arrival (Setup-1 case)



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Setup/Hold Time Illustrations

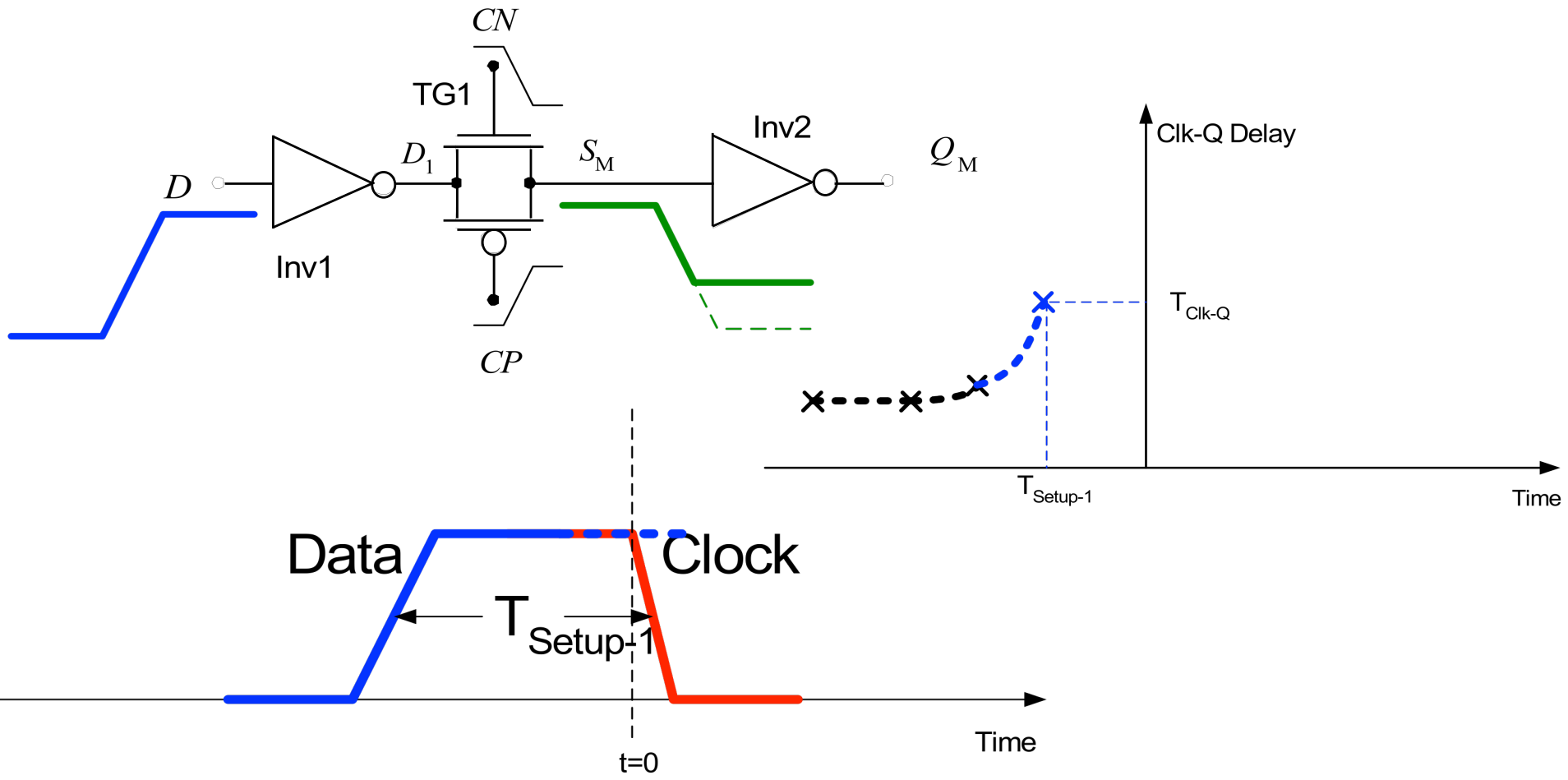
Circuit before clock arrival (Setup-1 case)



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Setup/Hold Time Illustrations

Circuit before clock arrival (Setup-1 case)



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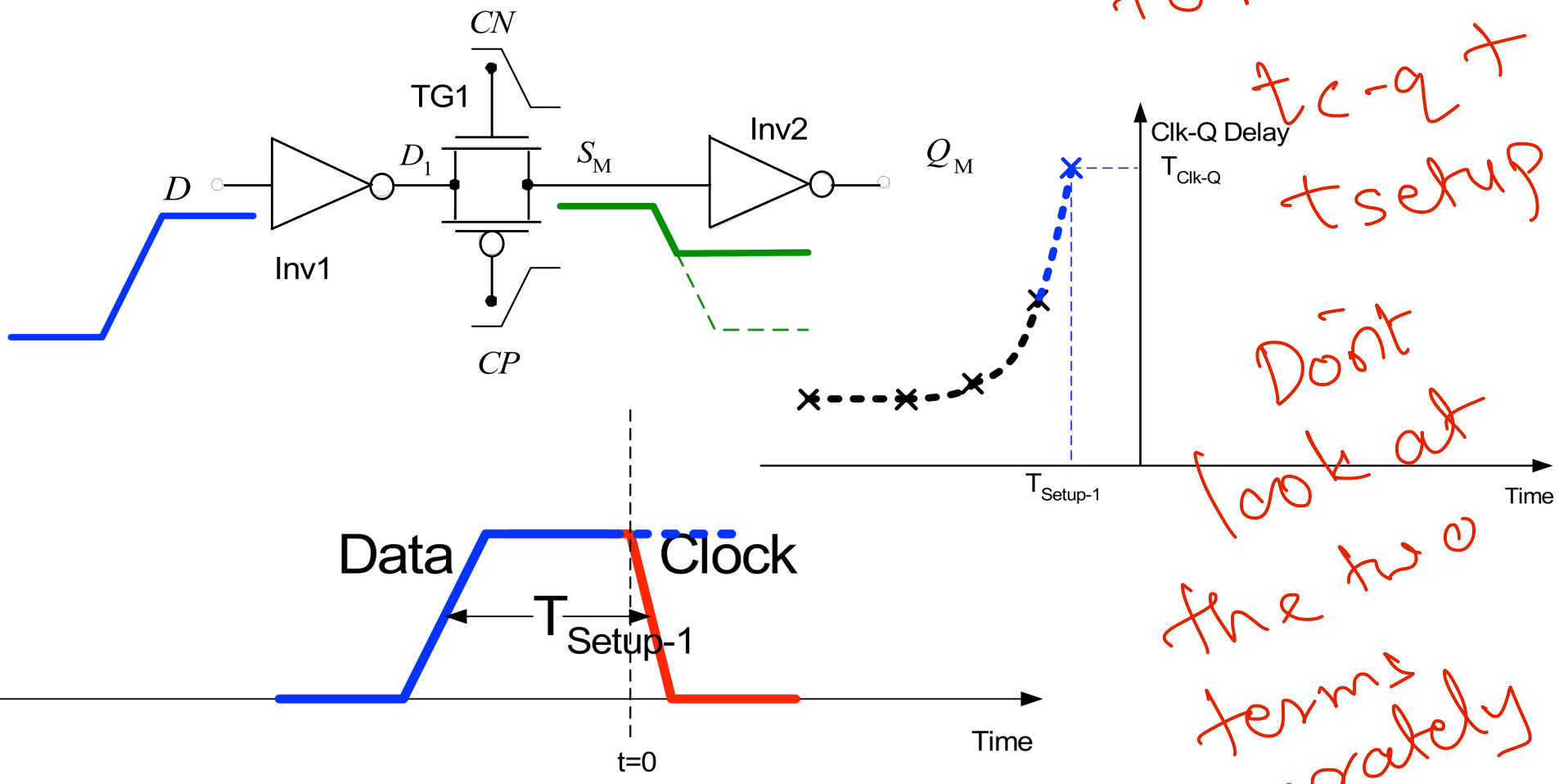
$t_{clk} \geq t_{c\text{-}q} + t_{logic_{longest}} + t_{setup}$

Setup/Hold Time Illustrations

We want to minimize $t_{c\text{-}q} + t_{setup}$

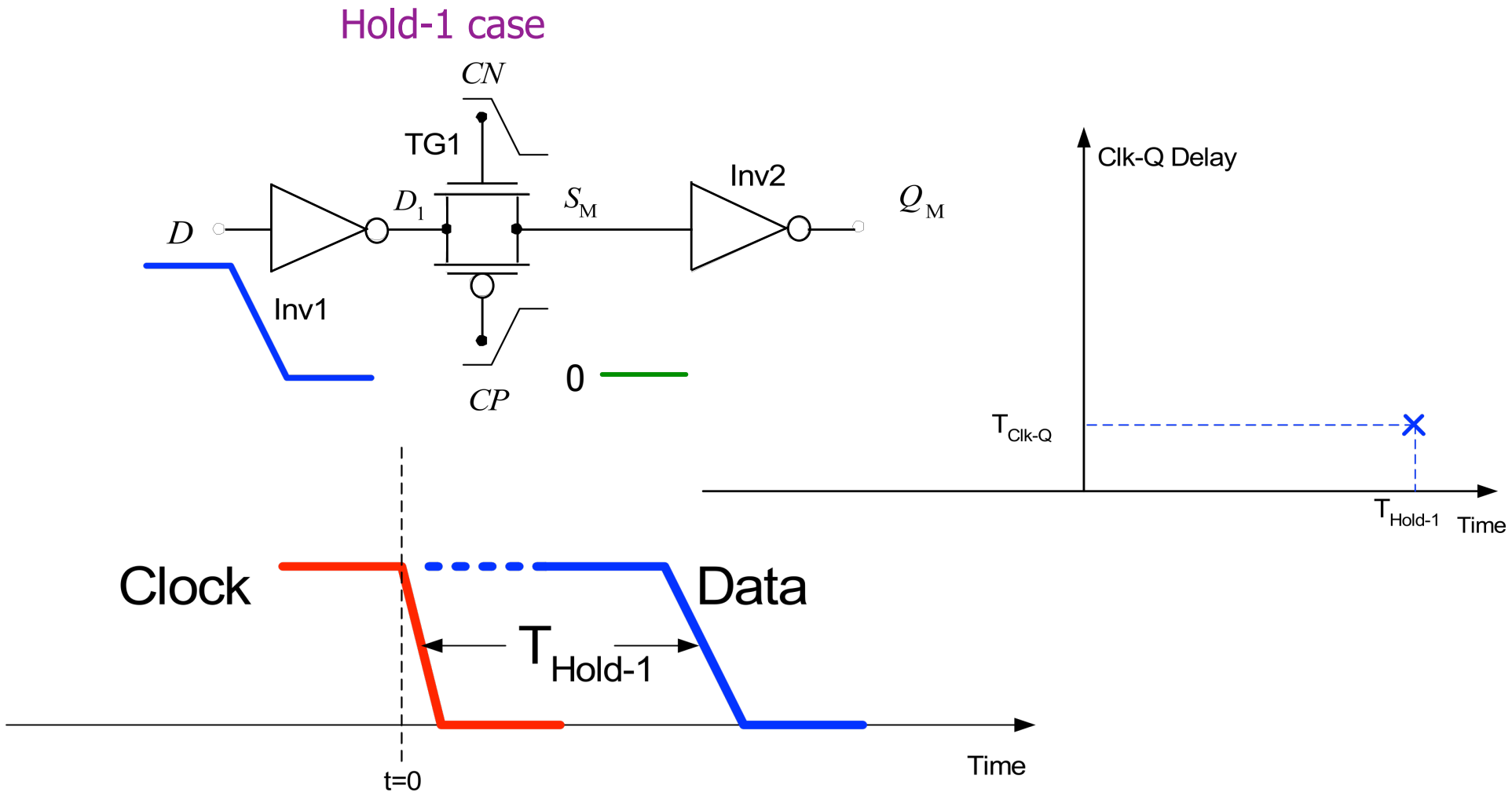
Don't look at the two terms separately

Circuit before clock arrival (Setup-1 case)



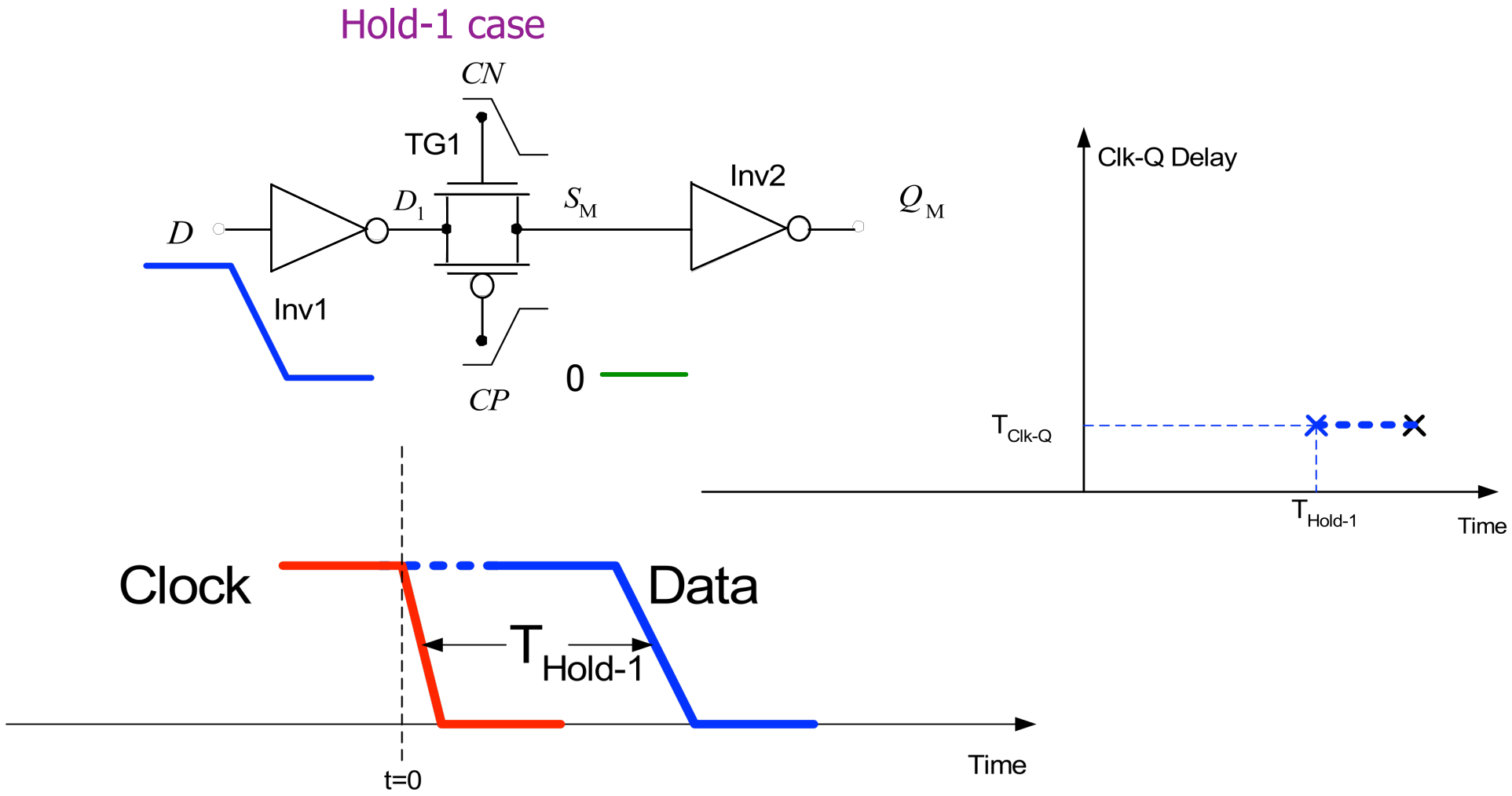
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Setup/Hold Time Illustrations



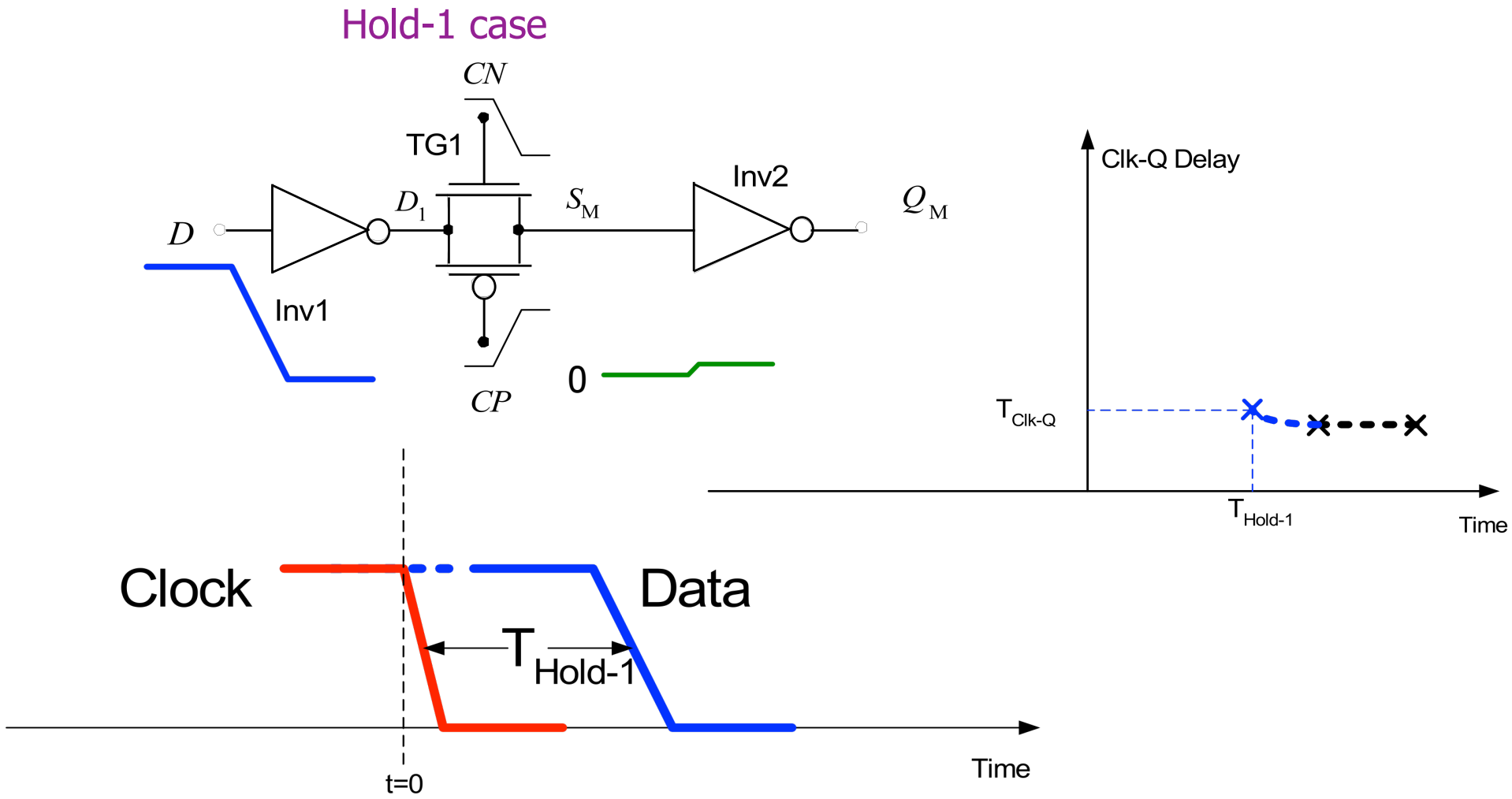
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Setup/Hold Time Illustrations



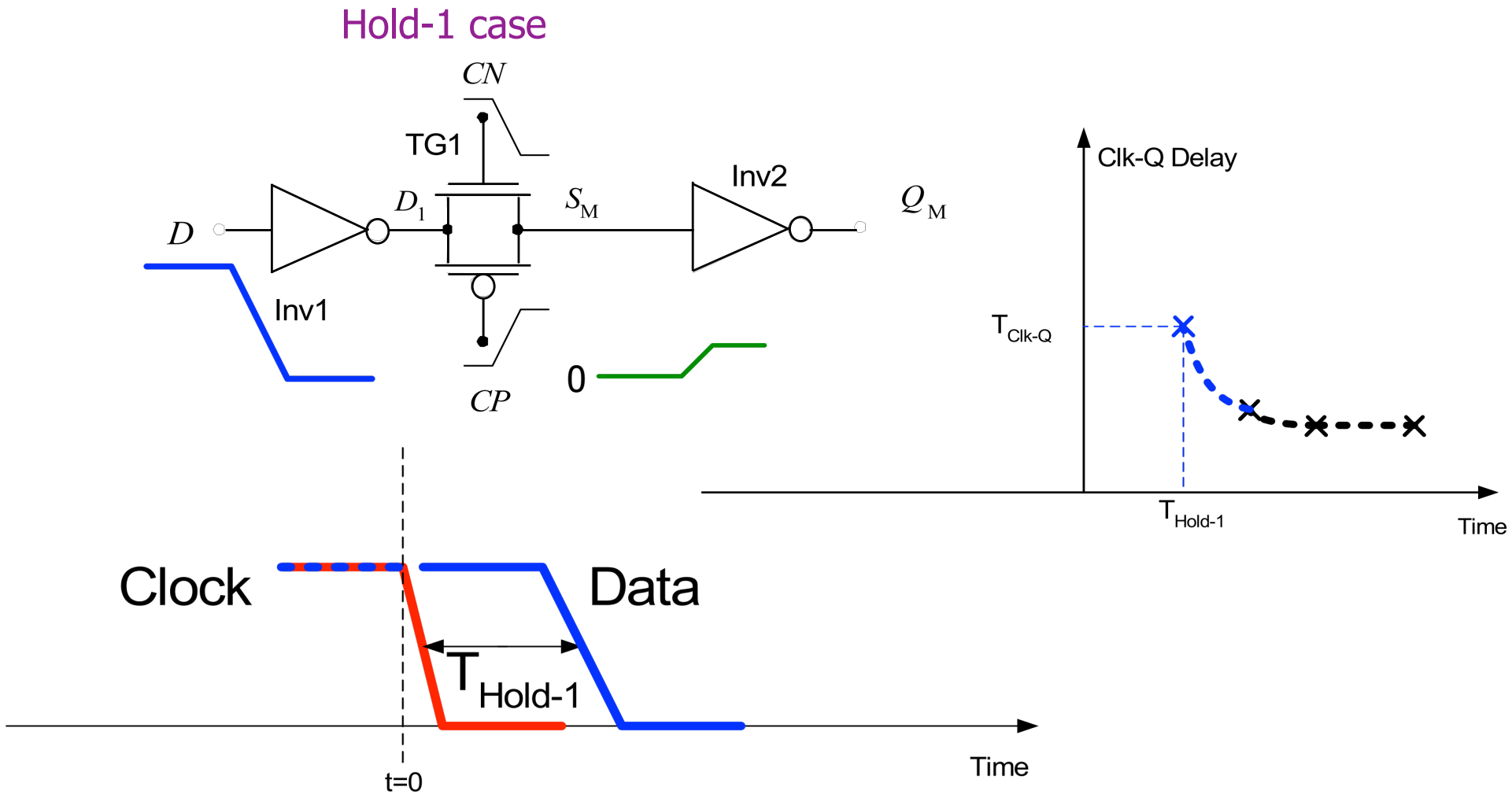
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Setup/Hold Time Illustrations



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Setup/Hold Time Illustrations

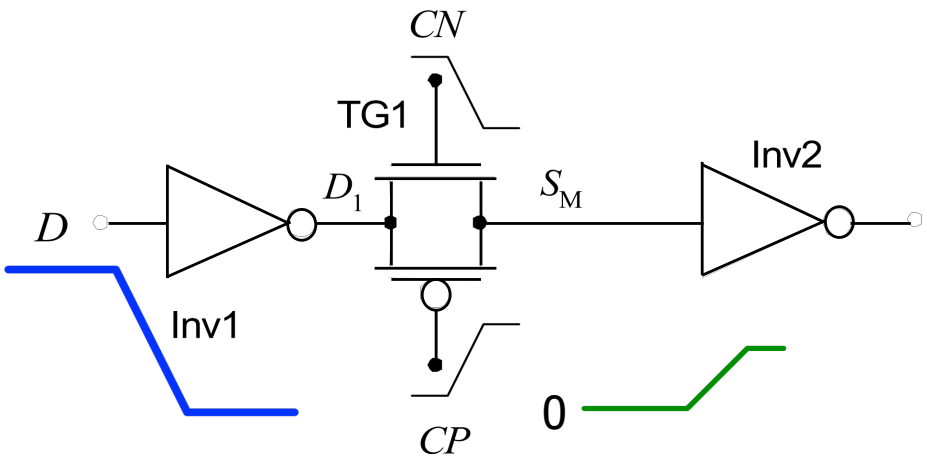


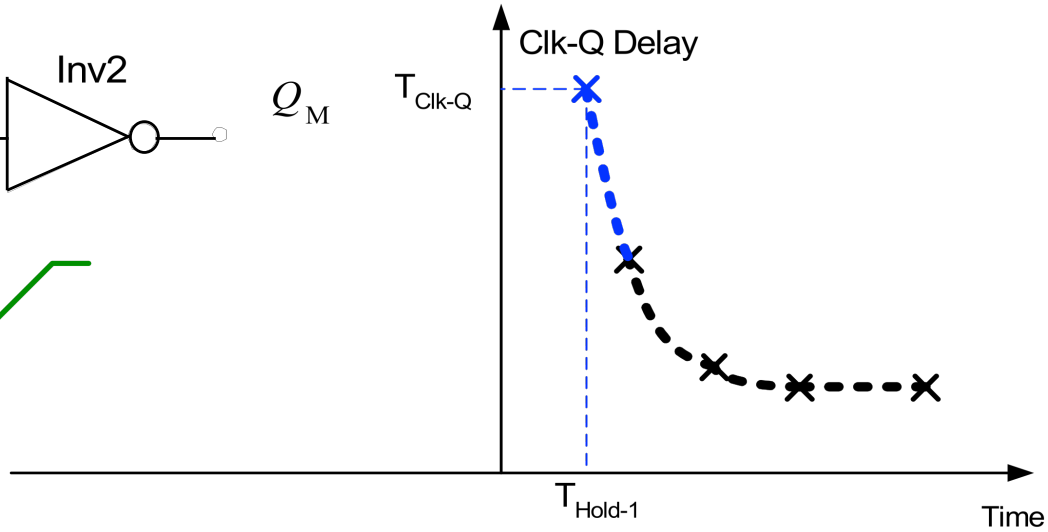
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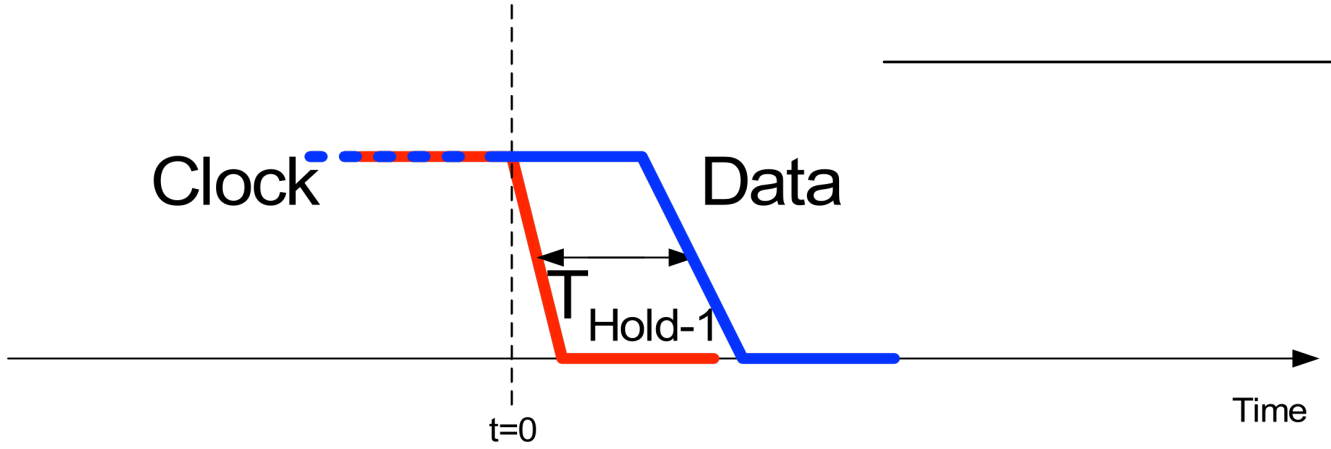
Setup/Hold Time Illustrations

$t_{hold} < t_{c\text{-}q} + t_{logic,\ shortest}$

Hold-1 case





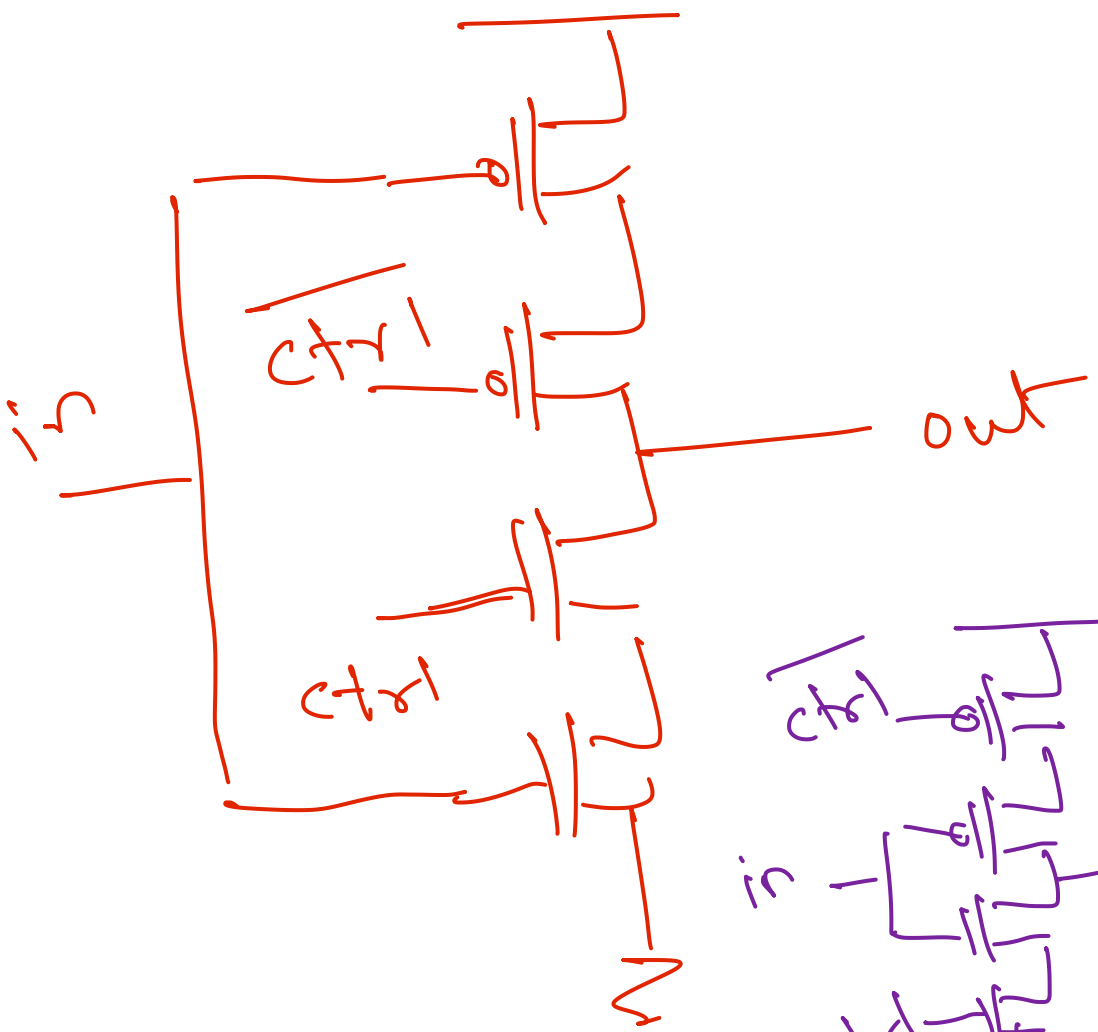


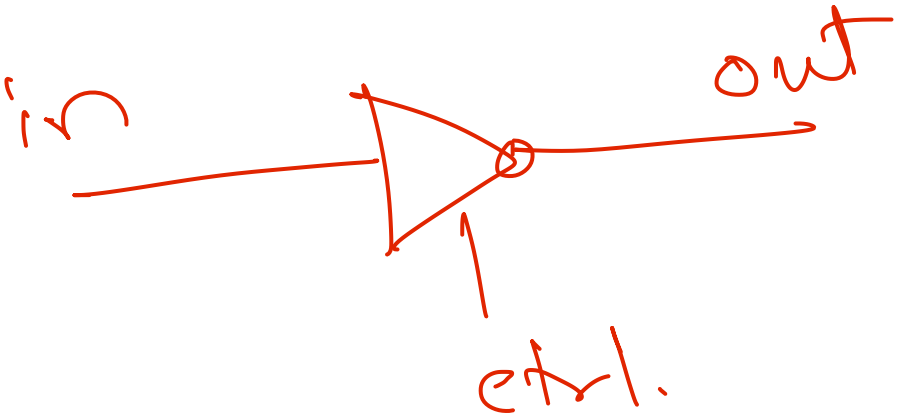
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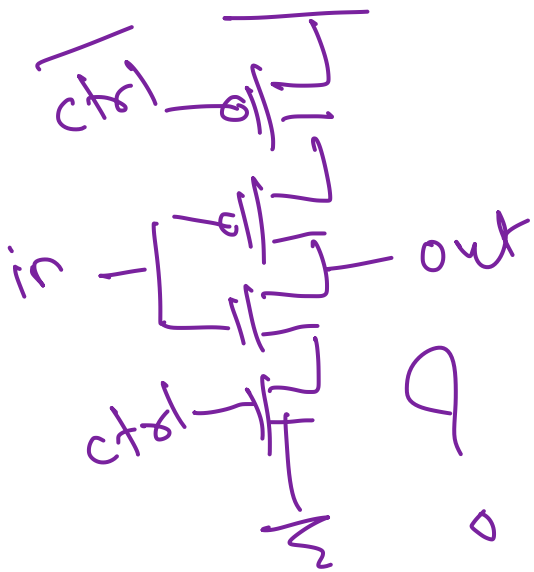
Other Latches / Registers

Using Tri-state buffers



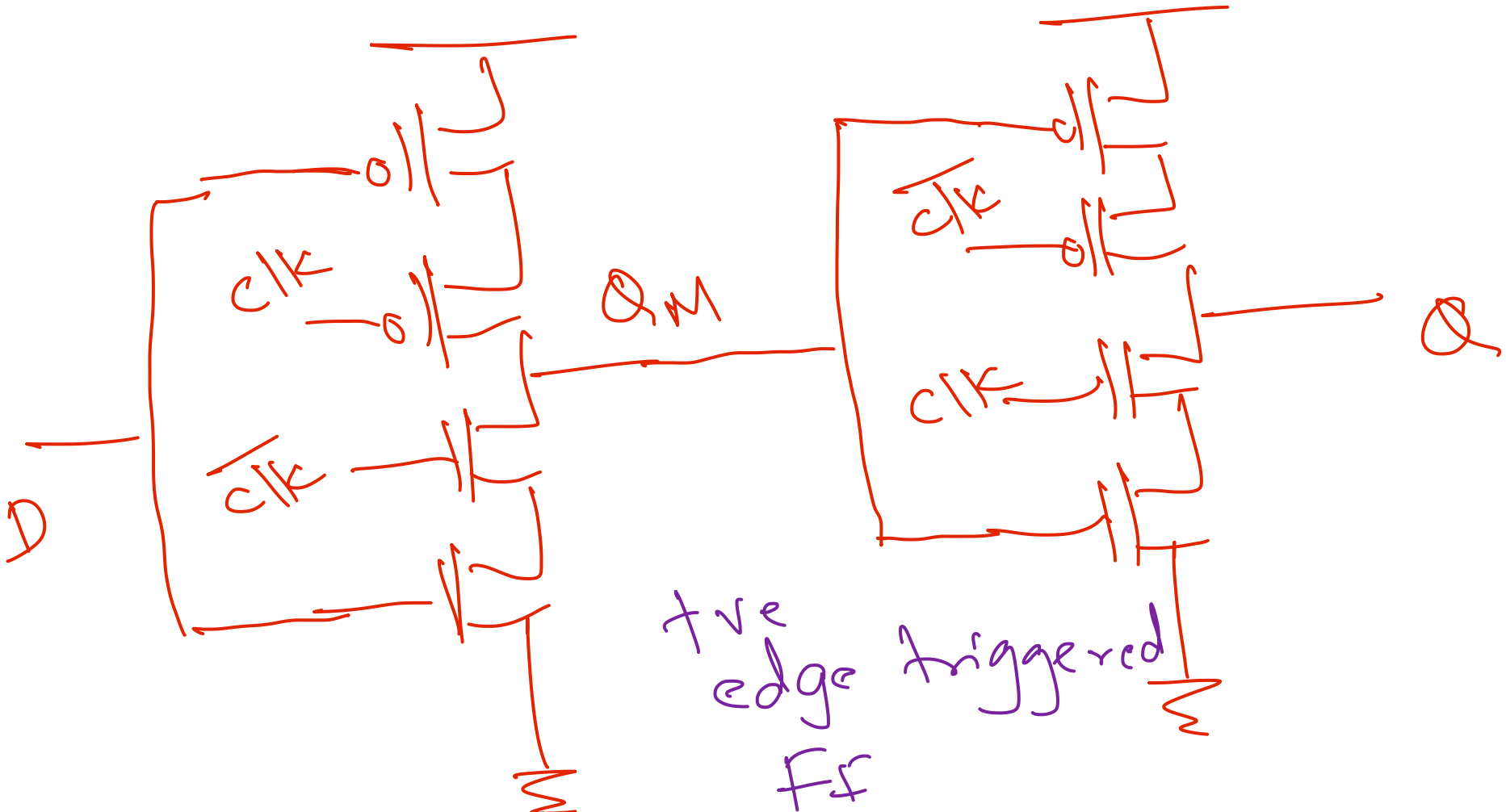


When
ctrl = 1
ckt. behaves like an inverter

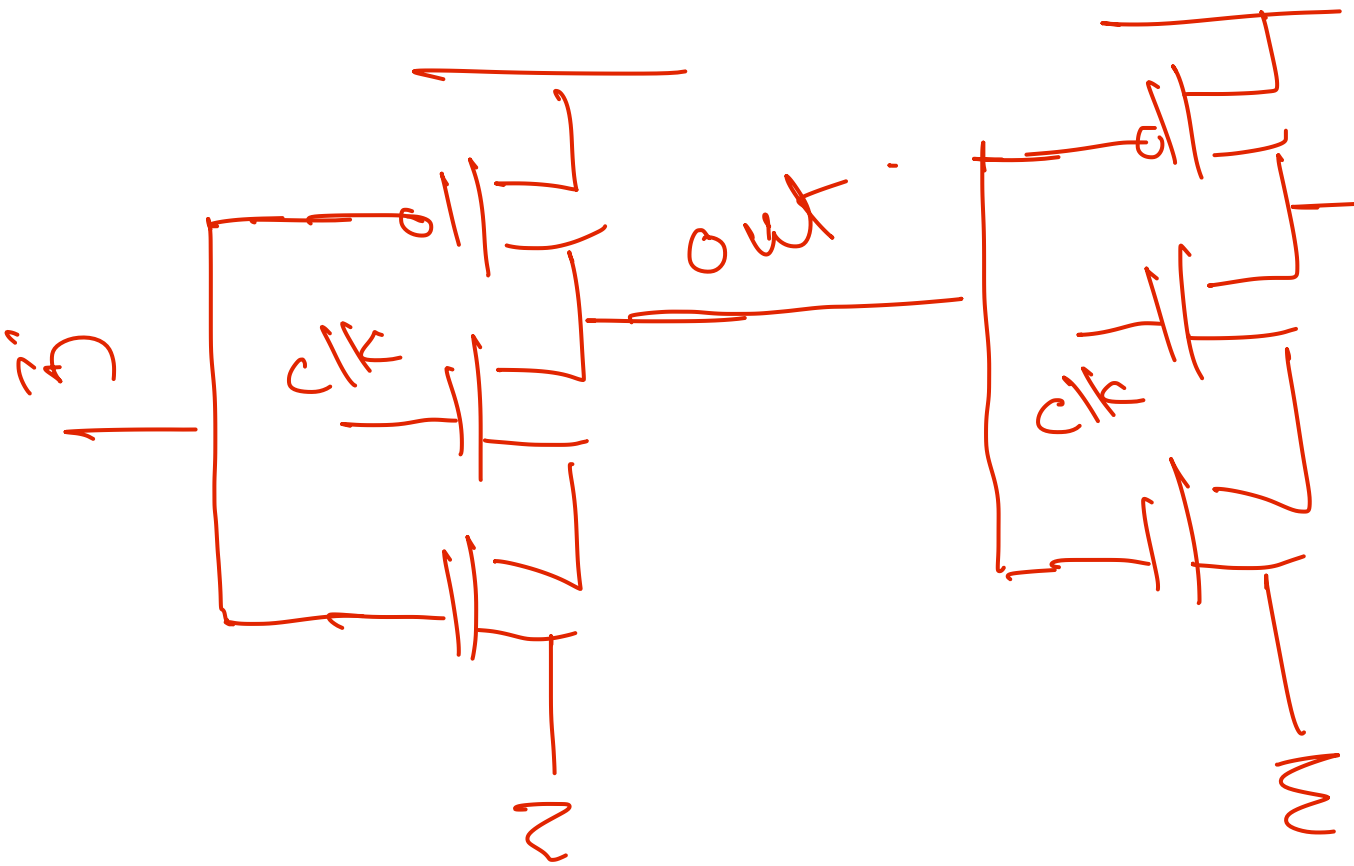


ctrl = 0
in is disconnected from out.

TS is effectively behaving like a +ve level triggered latch with inverted output

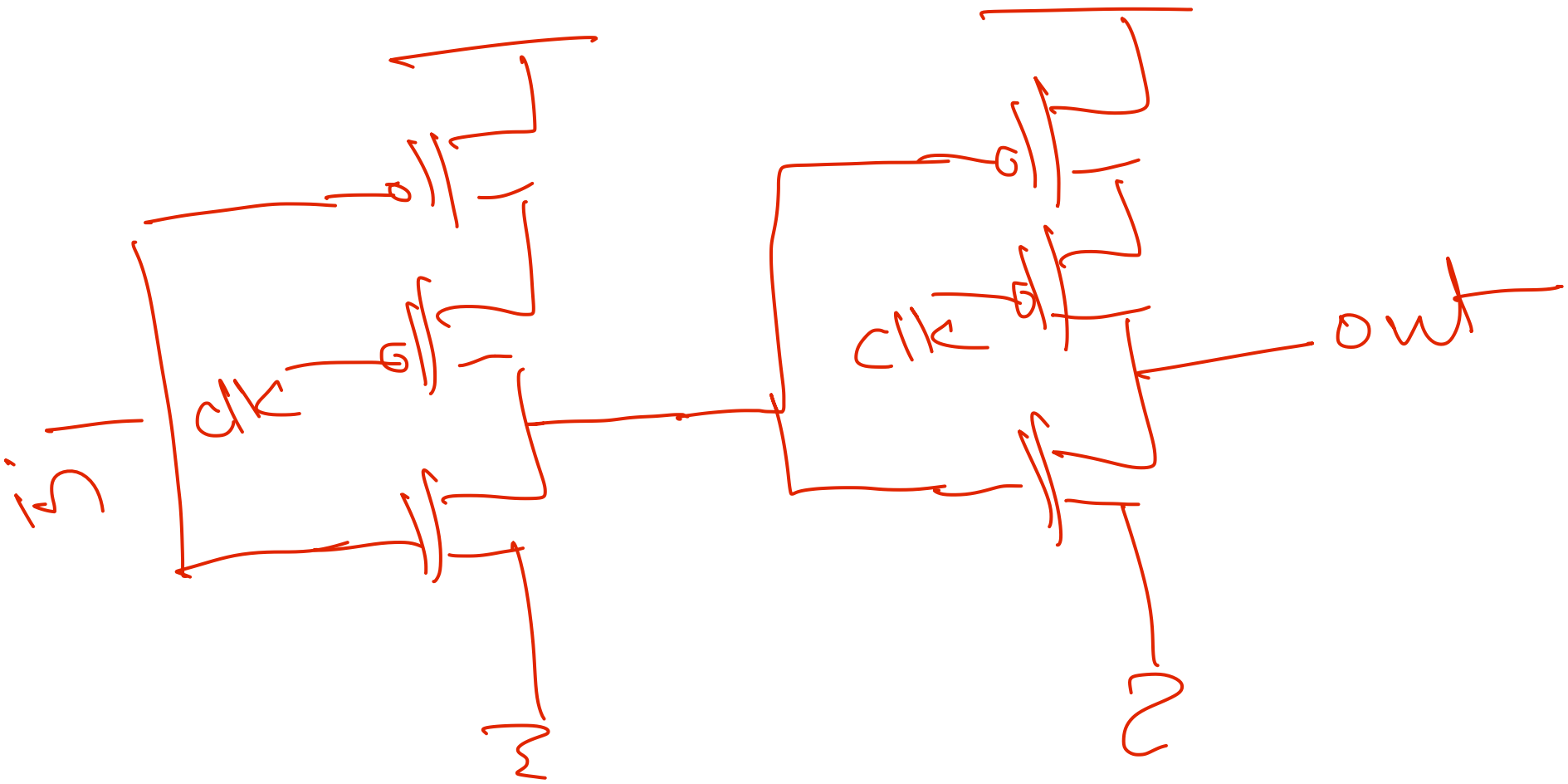


True Single Phase clk Latches

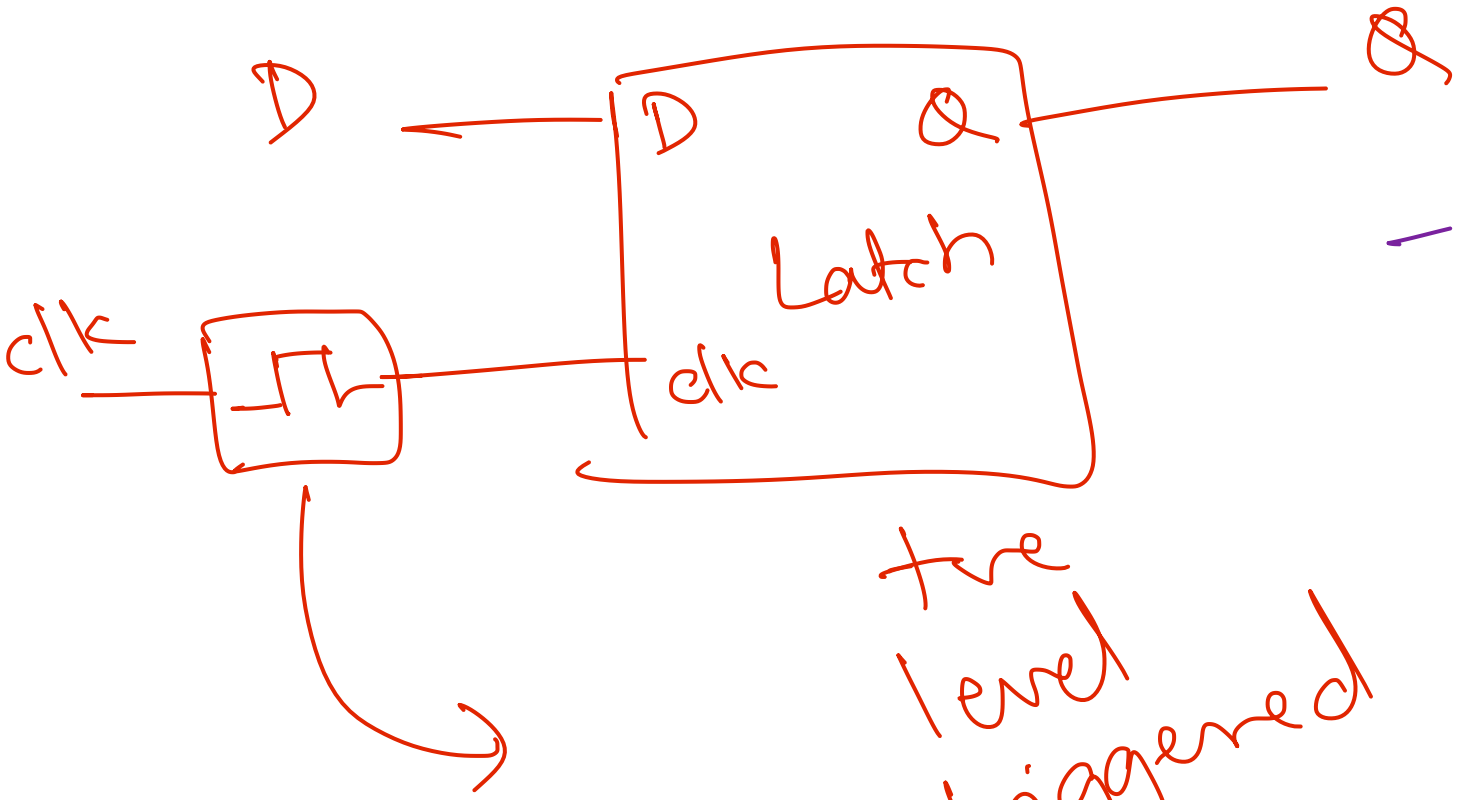


+ve level-triggered latch.

-ve level-triggered latch



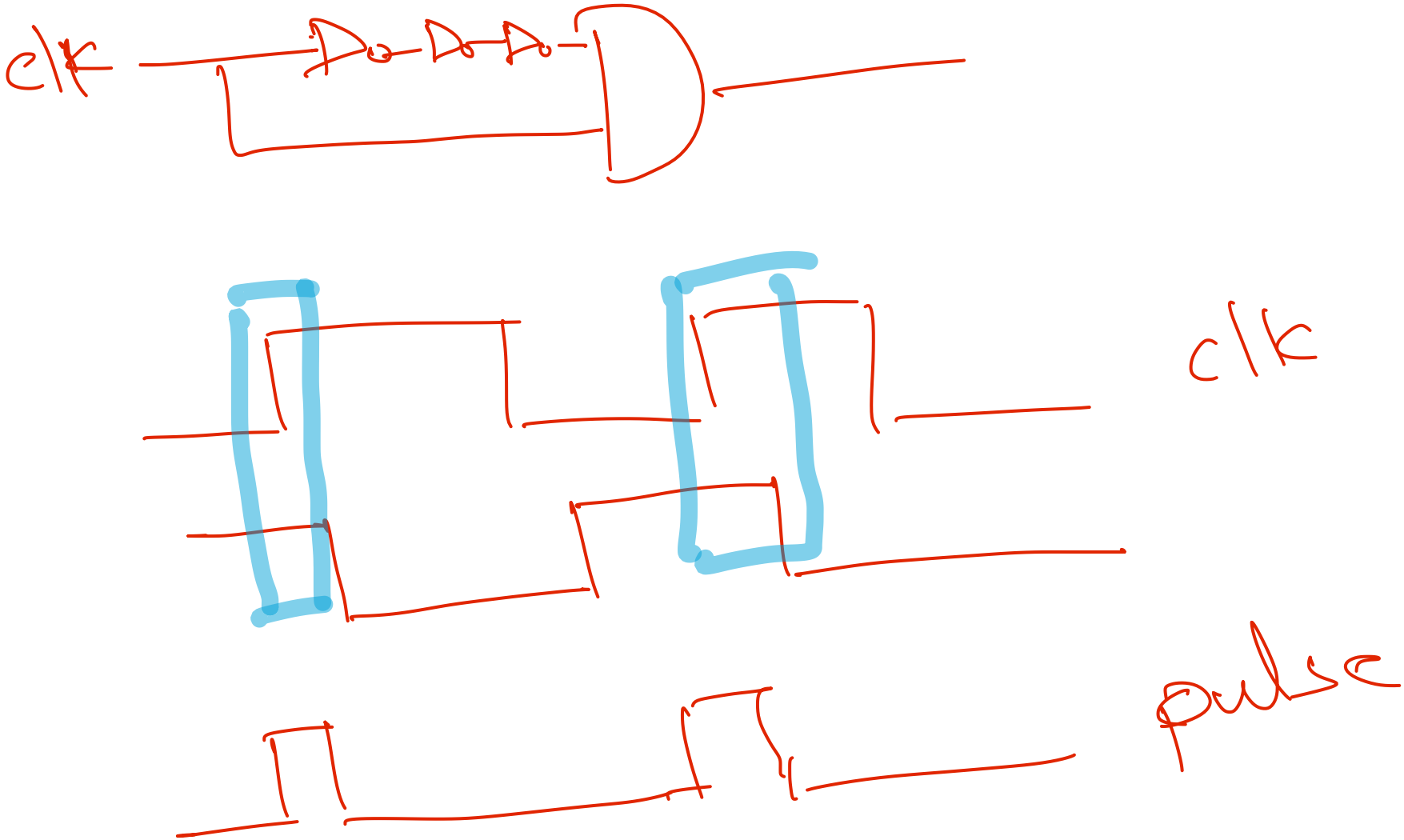
Pulse triggered latch



tve level triggered latch

- Behaves like a FF
- Cannot make the pulse small. Need sig. to pass through

- very


clk
clk
pulse